

BHANDARU PHANI PARASAR

Email: bhandarup@iisc.ac.in

Bangalore, India

EDUCATION

PhD Physics, Indian Institute of Science (IISc), Bangalore.

Jan 2023-

- Advisor: Prof. Vijay B. Shenoy, Department of Physics, IISc
- Cumulative GPA: 9.9/10

BS-MS (Research), Indian Institute of Science (IISc), Bangalore

2017-2022

- Physics major, Cumulative GPA: 9.2/10

RESEARCH PROJECTS

- Lieb-Schultz-Mattis theorem from gauge constraints** 2026
Demonstrated a novel mechanism to obtain an LSM-type theorem, wherein the symmetry responsible for the theorem originates from the kinematic constraints of a gauge theory. Illustrated this by constructing a $\mathbb{Z}_2 \times \mathbb{Z}_2$ gauge theory coupled to matter on a one-dimensional chain and studying the ground-state physics in the Gauss law subspace. It is shown that the theory in the Gauss law subspace has a U(1) symmetry whose generator commutes with lattice translations, but anticommutes with the lattice reflection operator. This leads to a Lieb-Schultz-Mattis (LSM) theorem at arbitrary fillings that rules out a trivial gapped ground state in the Gauss law subspace. Implementing the Gauss law is crucial for the existence of the LSM theorem. A gapless point, at which the excitations admit a description in terms of free Dirac fermions with a constraint on the total fermion number is identified, and the critical correlation functions are studied.
- Edge states of fracton systems** *in collaboration with* Yuval Gefen and Vijay B. Shenoy 2025
Explored the nature of edge states in fractonic systems and revealed its connection with the bulk properties, by studying the Chern-Simons theory of second rank gauge fields. Showed that braiding a quadrupole around a fractonic charge, and braiding two non-parallel dipoles produces a non-trivial statistical phase. Showed that there are two distinct edge modes—one fractonic and the other non-fractonic—and derived their current algebra. Studied the edge-to-edge tunneling and showed that it is a relevant perturbation.
- Delicate semimetals** *in collaboration with* Vijay B. Shenoy 2024
Proposed and constructed theoretical models for a novel class of semimetals, *Delicate semimetals*, which are protected by unstable homotopies. Our example of a two-band nodal-line semimetal in four dimensions is characterized by a Hopf flux. Studied their remarkable Fermi-arc and drumhead surface states and also generalized this construction using a three-dimensional example in the symmetry class AIII.
- Fermions coupled to $\mathbb{Z}_2$ gauge fields** *in collaboration with* Vijay B. Shenoy 2023
Studied fermions coupled to $\mathbb{Z}_2$ gauge fields on the square lattice and showed that a modulation of fermion hopping realizes of many Obstructed Atomic Insulators (OAIs). Analyzed the stability of these phases to quantum fluctuations of the gauge fields using mean field theory and found a rich phase diagram with many superfluids and features like Gross-Neveu criticality, triple points etc. Characterized the anomalous mass dimensions of topological and trivial masses at the Gross-Neveu critical point using renormalization group techniques.
- Phase transitions of fractons models** 2021
Bachelor's thesis supervisor: Prof. Vijay B. Shenoy
Worked on phase transitions of fracton models in three dimensions. Studied classical and quantum phase transitions in the X-cube and the checkerboard models using perturbation theory techniques and variational methods. Mapped the quantum partition function of the X-cube model to the classical partition function of a four-dimensional model, using trotterization.

PUBLICATIONS AND PREPRINTS

1. **Bhandaru Phani Parasar**, *Lieb-Schultz-Mattis theorem from gauge constraints*, arXiv:2605.13606 [cond-mat.str-el] (2026)
2. **Bhandaru Phani Parasar**, Yuval Gefen and Vijay B. Shenoy, *Bulk-Boundary Correspondence of Fractonic Field Theories*, Phys. Rev. Lett. 134, 236601 (2025)
3. **Bhandaru Phani Parasar**, Vijay B. Shenoy, *Delicate semimetals: Protected gapless phases from unstable homotopies*, Phys. Rev. B 109, 155131 (2024)
4. **Bhandaru Phani Parasar**, Vijay B. Shenoy, *Obstructed atomic insulators and superfluids of fermions coupled to $\mathbb{Z}_2$ gauge fields*, Phys. Rev. B 107, 245142 (2023)

WORKSHOPS AND CONFERENCES

- Presented my research at **APS Global Physics Summit**, Anaheim. March 2025
- Presented a poster at **A Hundred Years of Quantum Mechanics**, ICTS Bangalore. Jan 2025
- Presented a poster at **Young Investigators Meet on Quantum Condensed Matter Theory**, IISER Pune. Dec 2024
- Presented a poster at **Topological order:Anyons and Fractons**, Les Houches, France. April 2024
- Presented my research (virtually) at the **APS March meeting**. March 2024
- Presented a poster at **Condensed matter meets Quantum Information** at ICTS, Bangalore. Sep 2023
- Attended the workshop **Physics of Quantum Matter School** at NISER, Bhubaneswar. May 2023

FELLOWSHIPS

- **Prime Minister Research Fellowship (PMRF)**, instituted by the Ministry of Education, GoI 2023-
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** fellowship, instituted by DST, GoI 2017-2022

TEACHING

- PMRF teaching assistant for the NPTEL course *Statistical Mechanics*. 2026
Responsible for conducting problem solving sessions.
- Teaching assistant for *Solid state Physics*, *Mathematics for Physics I*, *Introductory Modern Physics*, *Electricity and Magnetism* and *Thermal Physics* at Azim Premji University, Bangalore. 2023-2025
Responsible for weekly interactions and tutorials.
- Teaching Assistant for *Quantum Mechanics II*, *Electricity and Magnetism* (lab) and *Topological phases of matter*, IISc. 2024-26
Responsible for taking tutorials, demonstrations (lab) and grading.
- Conducted problem solving sessions in physics at Azim Premji University. 2023
Responsible for mentoring undergraduate students.